

CMS Open Data H to tau tau Search: Phase 4b 10% Data Validation

Analysis my_analysis

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Change Log

Phase 4b v1

- Added deterministic 10% data validation using `(run * 1000003 + luminosityBlock * 9176 + event) % 10 == 0` on data rows only.
- Added the W high-mT control scale 0.8433 ± 0.0868 .
- Added fixed-model score-template validation plots for `vbf`, `boosted`, and `zero_jet`.
- The remaining 90% and full-data signal-region distributions remain blinded until Phase 4c approval.

1 Phase 4b Partial Inference: 10% Data Validation

1.1 Summary

Phase 4b validates the Phase 4a expected-primary score-template model on a deterministic 10% collision-data subsample. The fitted observable and category structure are unchanged from Phase 4a: `mva_score_hist_gradient_boosting` in `vbf`, `boosted`, and `zero_jet` channels with the Phase 4a score-bin edges. No model retuning, binning optimization, or replacement method was performed using the 10% data.

The 10% mask applies only to data rows and is exactly: `(run * 1000003 + luminosityBlock * 9176 + event) % 10 == 0`. MC templates use all selected MC rows but their official Open Data weights are scaled from $11.467/\text{fb}$ to $1.1467/\text{fb}$. The remaining 90% and full-data signal-region distributions are not inspected in this artifact.

1.2 W High-mT Control Scale

The W+jets high-mT control scale is derived from the masked 10% data as $(N_{\text{data_CR}} - \text{nonW_MC_CR}) / W_{\text{MC_CR}}$. The control region contains 400 10% data events, $W_{\text{MC}} = 241.478$, and $\text{nonW_MC} = 196.357$. The applied scale factor is 0.8433 ± 0.0868 with status `valid`. This factor is derived outside the signal region and is not tuned to improve the score-template agreement.

1.3 Score-Template Validation

Category	Data	Background	Data/background	Chi2/ndf	Max abs pull
vbf	9	14.64	0.615	2.690	3.24
boosted	228	205.92	1.107	0.387	0.96
zero_jet	883	688.35	1.283	2.121	2.01

Combined over all score-template bins, data/background is 1.232 with chi2/ndf 1.732. The score-modelling status is **flagged** under the predeclared Phase 4b gate.

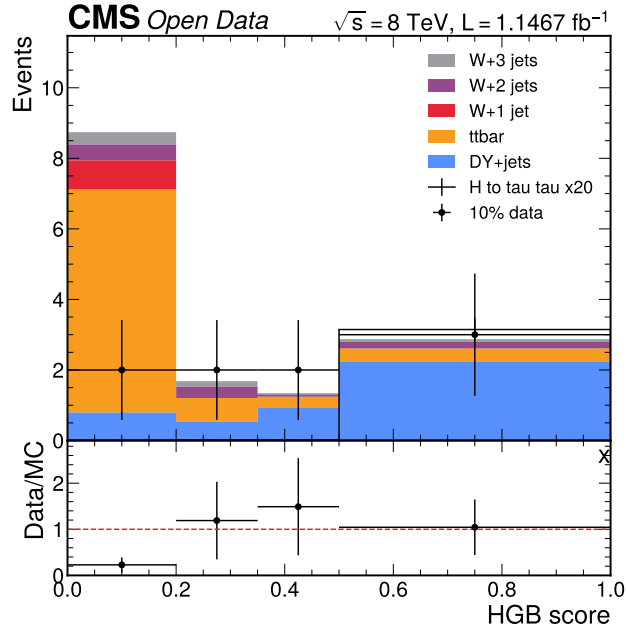


Figure 1: 10% score-template validation in the VBF category. The plot compares the masked 10% data score distribution to MC normalized to 10% luminosity with the W high-mT control scale applied. The ratio panel uses the same bins as the Phase 4a pyhf workspace and does not drive any retuning.

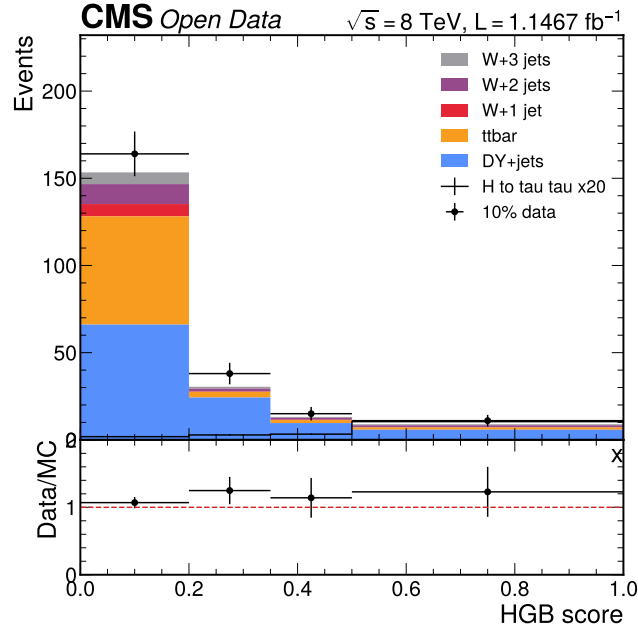


Figure 2: 10% score-template validation in the boosted category. The plot compares the masked 10% data score distribution to MC normalized to 10% luminosity with the W high-mT control scale applied. The ratio panel is a validation diagnostic for the fixed Phase 4a model.

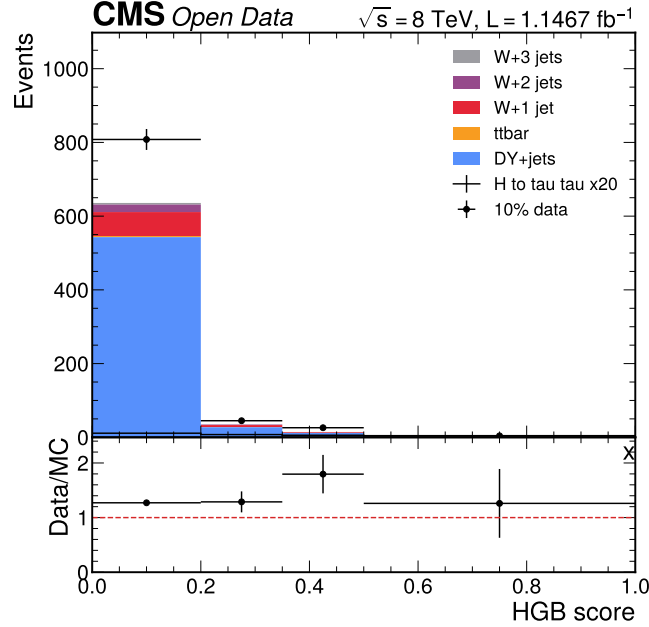


Figure 3: 10% score-template validation in the zero-jet category. The plot compares the masked 10% data score distribution to MC normalized to 10% luminosity with the W high-mT control scale applied. This category dominates the event count and therefore the combined ratio.

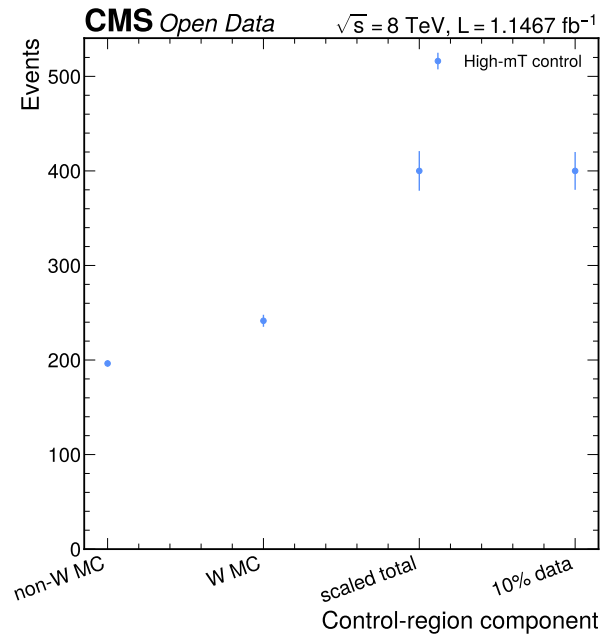


Figure 4: W high-mT control comparison. The figure shows the observed 10% data control count against the expected W and non-W MC components at 10% luminosity. The derived scale is propagated to W+jets in the partial workspace as a control region constraint.

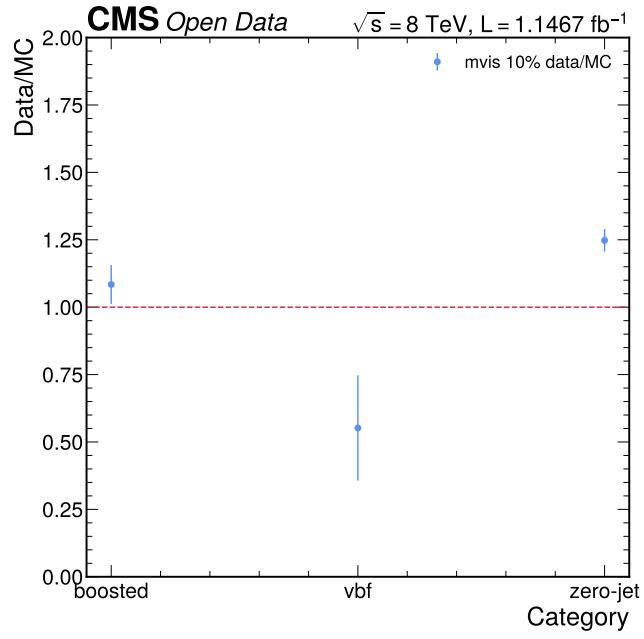


Figure 5: Visible-mass validation summary. The figure provides a mass-shape cross-check using the same masked 10% data and fixed categories. It is auxiliary to the score-template validation and is not used to choose a new observable.

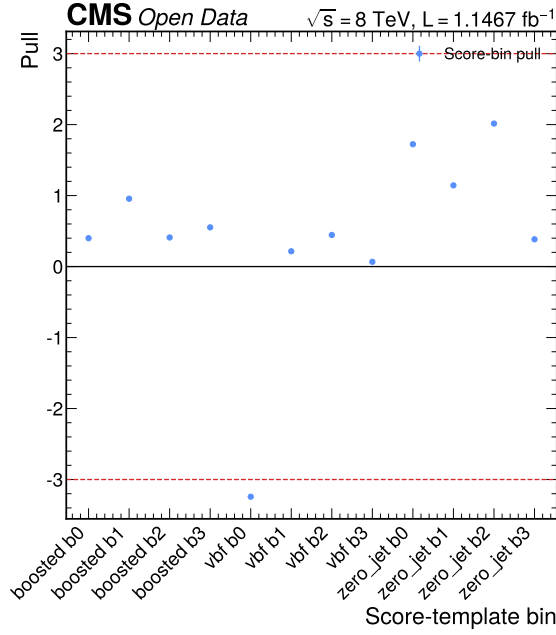


Figure 6: Score-validation pull summary. The figure summarizes per-bin pulls across the three score-template channels using the diagonal validation covariance. It highlights the largest observed 10% deviations without changing the model.

1.4 Partial Diagnostic Fit

The partial pyhf workspace in `pyhf_workspace_partial.json` uses the masked 10% signal-region observations. The diagnostic fit status is **evaluated**. If evaluated, its fitted signal strength is reported only as a 10%-data validation number and not as a final result.

1.5 Reproduction

Run:

Command	Purpose
<code>pixi run phase4b-results</code>	Build partial JSON, NPZ, workspace, and markdown outputs.
<code>pixi run phase4b-plots</code>	Render Phase 4b figures and compile the minimal 4b note.
<code>pixi run phase4b-validate</code>	Validate blinding metadata, workspace construction, and figures.
<code>pixi run phase4b-all</code>	Run the full Phase 4b executor chain.

2 Phase 4a Context

The Phase 4a expected model is the simultaneous pyhf score-template fit in **vbf**, **boosted**, and **zero_jet** channels using `mva_score_hist_gradient_boosting` with bin edges `[0.0, 0.2, 0.35, 0.5, 1.0]`. Phase 4a reported a median expected 95% CLs upper limit of 4.199 and an expected discovery diagnostic of $Z = 0.526$.

3 Unblinding Checklist

Gate item	Phase 4b status
Background model validated on 10% data	flagged
Systematics retained	luminosity 2.6%, DY 15%, tau/open-data 15%, W high-mT control
Expected result physically sensible	inherited from Phase 4a expected artifacts
Signal injection/closure	inherited from Phase 4a expected artifacts
10% partial unblinding pathologies	see score-template and W high-mT validation summaries
Remaining data blinded	yes

References